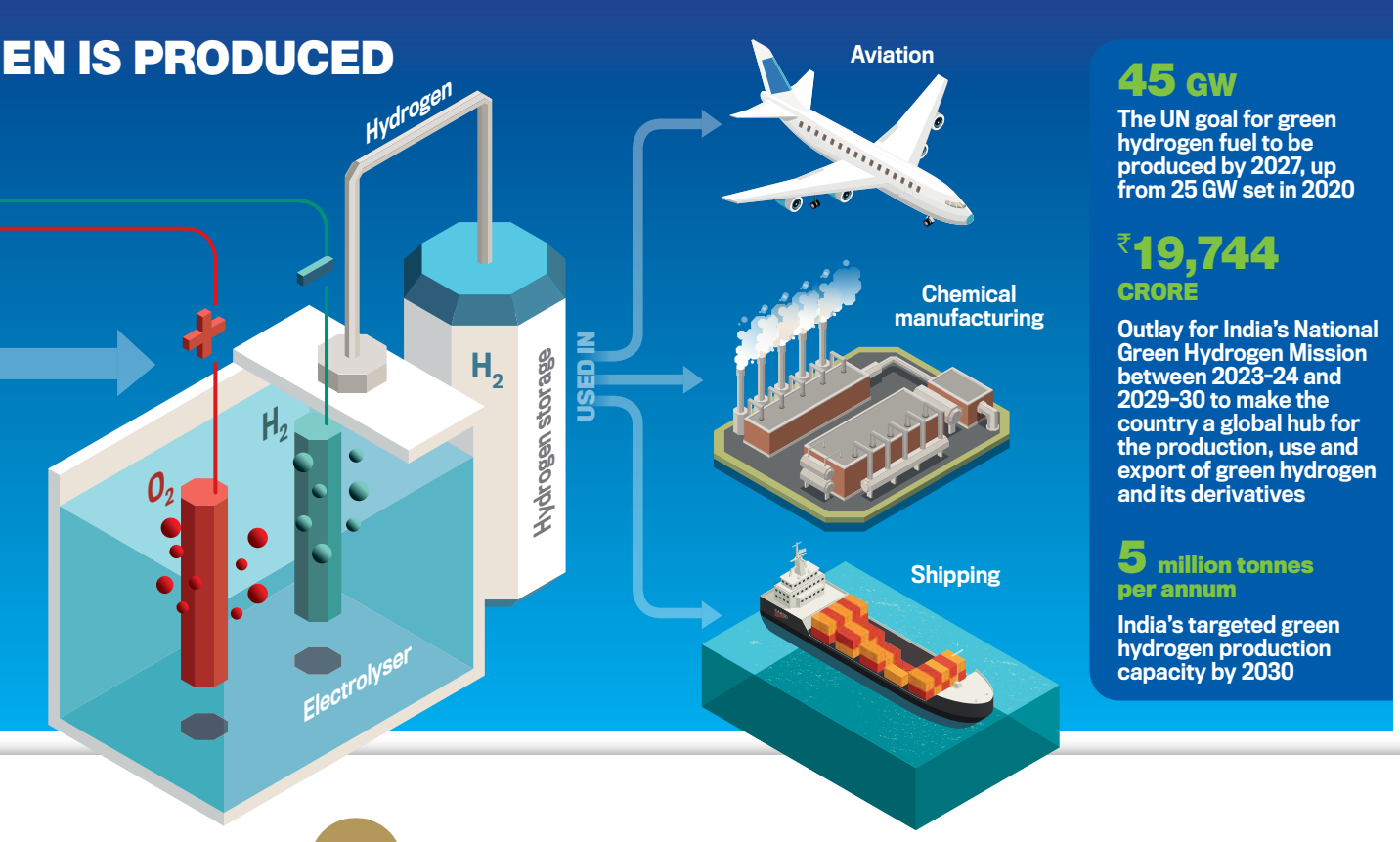


45 GW
The UN goal for green hydrogen fuel to be produced by 2027, up from 25 GW set in 2020

₹19,744 CRORE
Outlay for India's National Green Hydrogen Mission between 2023-24 and 2029-30 to make the country a global hub for the production, use and export of green hydrogen and its derivatives

5 million tonnes per annum
India's targeted green hydrogen production capacity by 2030

lakh crore) investment in 10 years

- The strategy will comprise three business streams: making supply chain products; generating green hydrogen; producing downstream derivative products

RELIANCE INDUSTRIES

- Is reportedly investing Rs 6 lakh crore in multiple green energy projects in Gujarat as part of its ambitious decarbonisation drive
- Will invest Rs 5 lakh crore over 10 to 15 years to set up a 100 GW renewable-energy power plant and develop a green hydrogen ecosystem. It is said to be scouting for land for the projects in Kutch, Banaskantha and Dholera in Gujarat
- Will spend an additional Rs 60,000 crore in setting up 'new energy manufacturing', which includes solar modules, electrolysers, batteries for energy storage and fuel cells

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- Has entered into a technology licence pact with French firm McPhy-Energy to manufacture electrolysers, including future product upgrades
- Has formed a joint venture with Indian Oil Corporation and ReNew for green hydrogen
- Will provide integrated EPC (engineering, procurement and construction) and modular fabrication services for the emerging green hydrogen and ammonia segments
- Has commissioned India's first green hydrogen plant at its Hazira manufacturing complex with a capacity of 15 tonnes per annum

$158.3 billion (Rs 13 lakh crore). As the world increasingly pledges to move away from fossil fuels, India will be looking to cut down on oil imports and bank on indigenous production of green fuel in the long run.

WHAT INDIA NEEDS TO MASTER

In sectors that are known for hard-to-abate emissions—that are either prohibitively costly or impossible to reduce with currently available abatement technology, including steel, cement and oil refining—India needs to start making the shift to using renewable energy. For instance, the only way to make an oil refinery that uses hydrogen as an input clean or green is to replace grey hydrogen with green hydrogen.

This presents an opportunity for companies to enter green hydrogen production initially for industrial use and, as the technology evolves, to use it in hydrogen-powered vehicles. Meanwhile, countries such as the US and several European nations are making the transition to green hydrogen. This has led to large-scale requirements of electrolysers, offering a window for Indian heavy engineering firms to step into their manufacture for the global market. For this, more firms should be encouraged to tie up with global giants for the skill and technology to make modern machinery or even green hydrogen and green ammonia plants. That is the way India can go green and grow. ■